

Bonus task 2

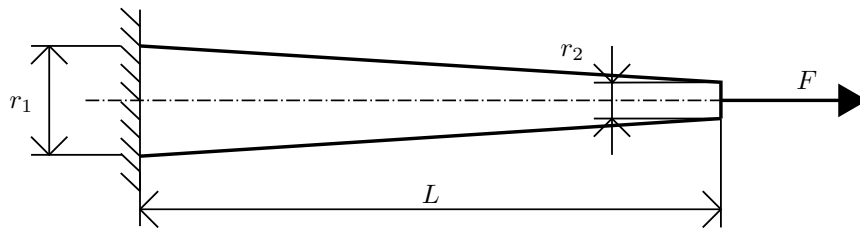
Rod with variable cross-section.

Possibility to earn five bonus points

Deadline: Before the lecture on 3th of June

The aim is to calculate the displacements at the free end of a tapered rod structure using a **linear two-node finite rod element** in MATLAB. The length of the structure is $L = 1000$ mm, with a circular cross-section radius of $r_1 = 50$ mm at the fixed end and $r_2 = 25$ mm at the free end. Young's modulus is $E = 210000$ MPa, and an applied force of $F = 1000$ N is exerted at the free end. Discretise the problem using three different meshes with 2, 4, and 8 elements, and solve for the displacements. Compare the FE results with the analytical solution.

Write a brief report that includes the node and element numbering, boundary conditions, and the approach to discretising a structure with a variable (tapered) cross-section. You must use the shared MATLAB templates to solve the problem. Bonus points will be awarded as follows: correct global nodal displacements at the free end (3 points, 1 point per mesh), a clear report including a comparison between FEM and the analytical solution (1 point), and well-commented MATLAB code (1 point). You must submit the MATLAB code (m-files) and a report of approximately two pages. Please note that no points will be awarded without both the report (in PDF format) and the MATLAB code (m-files).



TIPS

Use code templates `run_static_TwoRodElements.m` and `KlocRod.m` as a starting point. Note that you do not need to transform a local elemental coordinate system to a global coordinate system because the rod structure is straight and parallel with the global coordinate system.

Cross-section varies linearly, so it can be expressed as follows:

$$A(x) = A_1 \left(1 - \frac{x}{L}\right) + A_1 \frac{x}{L}$$

Analytical solution

The axial displacement of the linearly tapered rod can be calculated using the following equation:

$$u(L) = \frac{FL}{E(A_2 - A_1)} \ln \left(\frac{A_2}{A_1} \right)$$

Comparison of FEA and analytical displacement solutions

Your FE-solutions should be something like that:

